Marcus Turner-Wood

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Personal Statement

PhD student in experimental nuclear physics at the University of Liverpool, working on single-neutron transfer reactions at the ISOLDE Solenoidal Spectrometer, CERN. Founder of the Chadwick Institute Young Nuclear Network and of the University of Liverpool Nuclear Society, which I affiliated with the Nuclear Institute while helping to build a UK-wide network of nuclear societies. Experienced in data analysis, simulation and machine learning, including industry projects on digital twins and AI agents during an internship at VEC. Selected for the Liverpool Leadership and Enterprise Academy and recognised for initiative, motivation and a proactive attitude to learning.

Key Skills

- Python, C++ and simulation codes; machine learning, deep learning and data analysis
- Full-stack development (FastAPI, SQLite, JavaScript), simulation and digital twins, agentic AI
- Gamma-ray and charged-particle spectroscopy, detector modelling and uncertainty analysis
- Leadership, project organisation, teaching and public speaking
- Strong networking skills, particularly within the nuclear industry

Education

University of Liverpool **10/2026 – Present**
PhD in Nuclear Physics
Supervised by Prof Robert Page and Dr Liam Gaffney. Studying single-particle structure near the lead isotopes through (d,p) transfer reactions at the ISOLDE Solenoidal Spectrometer (experiment IS753, ^{196}Pb(d,p)^{197}Pb), including Doppler correction, efficiency and Geant4 modelling of the $CeBr_3$ gamma-ray array, and analysis of the LUME luminosity monitors. Taking the taught CPD modules PHYS804, PHYS806, PHYS808 (Nuclear Instrumentation) and PHYS816, alongside the departmental C++ course.

University of Liverpool **09/2023 – 06/2026**
BSc Physics with Nuclear Science
Final-year project: *Gamma-gamma coincidence spectroscopy of ^{175}Au: evidence for shape coexistence and a machine-learning coincidence interpreter.* Analysed recoil-decay tagged data from the JUROGAM, RITU and GREAT set-up at the University of Jyväskylä Accelerator Laboratory to build an updated level scheme, identifying weak linking transitions and band structures of different deformation through moment-of-inertia systematics. Developed a reproducible coincidence-analysis workflow and a machine-learning interpreter that sets the parameters of the coincidence analysis and returns a peak list, a ranked coincidence graph and candidate cascades.

Other projects completed in an academic setting:

- Radioactive waste classification, non-destructive assay and the role of digital twins: HPGe and NaI(Tl) measurements on a laboratory phantom, Monte Carlo attenuation modelling and a digital-twin prototype for waste classification
- Radio-isotopes in fusion–fission hybrid reactors: activation and shielding studies with gold and vanadium foils and LabSOCS efficiency modelling
- Machine learning for FTIR histopathology: CNNs, SVMs, hierarchical clustering and Kolmogorov–Arnold networks for tumour detection in biopsy data
- Pressure-sensitive bed sheet for falls and pressure-ulcer management in hospitals, with a machine-learning predictor based on position, sleep stage and weight distribution

Further details at marcusturnerwood.com/projects.

Liverpool Leadership and Enterprise Academy **01/2025 – Present**
Selected student
Earned a place as one of 20 students selected from 180 applicants across the science and engineering departments, based on previous e-commerce experience and demonstrated leadership.

Work and Teaching Experience

Teaching Assistant, Undergraduate Laboratories **10/2026 – Present**
University of Liverpool, Department of Physics
Demonstrating the Zeeman effect experiment to undergraduate students: guiding the optical set-up and measurements, explaining the underlying atomic physics, and supporting students with data analysis and write-ups.

Intern in Data Analysis and AI **06/2026 – 10/2026**
VEC
Projects covering data analysis, full-stack development, agentic AI and digital twins:

- UK spin-out tracker: an interactive dashboard comparing university spin-out activity with research funding, built on HESA, UKRI and Companies House data
- Embroidery production digital twin (Wild Thang): a live twin of a production line with NFC operator tracking, machine production data, an integrated AI assistant and computer-vision research into factory-floor tracking
- Chester Racecourse crowd and queue simulation: a race-day digital twin simulating crowds, queues and spending on the site map, with an operations dashboard, incident prediction and a georeferenced 3D model of the estate
- Agentic factory: tested whether LLM agents can outperform rule-based production control, through two simulated comparison studies, a proposed agentic simulation platform and a customer-facing garment design demonstrator

Research of Plastic Scintillators with Medical Applications **06/2025**
Sapienza University of Rome
Optimised the threshold voltage, discriminator width and PMT voltage of a two-detector plastic-scintillator cosmic-ray telescope, using chi-squared fits and machine learning (CNNs, SVR, random forests) to map the parameter space. Presented the work to a group of researchers and PhD students.

Quality Control Lab Technician **Summer 2024**
Thatcher's Cider
Conducted tests and assessments on the quality of fruit, juice, and aromas during the blackberry season. Performed an average of ten tests per hour and reduced hourly test times by over 30% using decision analysis. Completed 12-hour day and night shifts while consistently arriving early or on time.

Leadership and Additional Experience

Founder, Chadwick Institute Young Nuclear Network
Founded the Chadwick Institute Young Nuclear Network, a network connecting nuclear researchers at the Universities of **Liverpool** with other student groups, with industry and membership bodies. Wrote a number of data-backed research papers demonstrating the group's effectiveness and how best to operate.

President and Founder, Liverpool Nuclear Society
Founded the society to bring together students interested in nuclear science and the nuclear industry. Organised guest lectures, presentation evenings, and career-focused activities, and affiliated the University of Liverpool with the Nuclear Institute. Worked closely with nuclear societies at other universities and with industry, building a large and growing network of academics and industry professionals; through these connections the society secured an industry internship for one of its members.

Nuclear Institute Educational Affiliate
Attended Young Generation Network talks and Nuclear Institute branch formal dinners, contributed to volunteer outreach, and supported country-wide networking between nuclear societies. Attended all of these alone with the purpose of meeting new and like-minded people.

Mentoring Scheme
Set up a sixth form mentoring scheme involving 28 students, providing academic support to younger and struggling students. Used decision analysis to organise the project and successfully maintained mentorships for 26 of the 28 applicants. The programme was independently conceived and backed by middle and senior staff.

Gold Duke of Edinburgh Award
Took a leadership role during expeditions, including planning routes, campsites, emergency escape routes, and support for less able team members. Demonstrated calm decision-making and confidence in challenging situations.

Personal Projects

- ISOLDE Explorer and mock digital twin: an interactive 3D model of the ISOLDE facility at CERN built from CERN CAD, following the beam from production and separation through charge breeding and acceleration to the experimental stations (ISS, Miniball, SpecMAT and others), with live simulated detector events. A mock digital-twin overlay proposes measurement points across the facility and shows synthetic live readings, trends and fault scenarios (beam trip, RF drift, missing sensor), as a basis for discussing integration with real data sources.
- Research organiser: a self-hosted web application (FastAPI, Python, SQLite) for managing academic reading, with a PDF library, citation extraction, analytics, a research diary, code snippets and a research feed.
- NeuWave ship tracking: a Python tool that decodes AIS vessel broadcasts (speed, course, heading, anchor state) into a live, clustered map. Also developed six online e-commerce businesses and received mentorship from an industry leader in the party and concierge sector.
- Stellarator plasma modelling: equilibrium modelling with VMEC (Variational Moments Equilibrium Code) and SPFINCS (Stellarator Plank-Fokker Iterative NeoClassical Solver), towards a physics-informed neural network predicting confinement times from operating conditions and stellarator geometry.
- Blog of research articles, written in HTML, on topics including Amanita muscaria, Lucretius (physics, philosophy, poetic form and influences), Epicureanism in the modern day, the effects of UKRI funding cuts, Dalí's scientific influences, materials for fusion, the economics of STEP, the physics of stellarators, and Europe's fusion reactors. Website: `https://marcusturnerwood.com/articles/`.